to revenues provides an important insight on the capacity to service public debt. Sri Lanka's debt- to-revenue ration has continued to increase in recent years:

- Total Debt/Revenue Ratio- 2012 - 571%, 2013 - 597%,2014 - 618%
- The debt servicing cost as a proportion of revenue is a better indicator of the crippling effect of the large public debt.
- Total debt service /Government Revenue 2011 - 92.5% , 2012 - 96.8% , 2013 - 102.2%
- In 2012, debt servicing payments absorbed 96.8% of revenue, In 2013, debt servicing costs were higher than the government revenue- 102.2% of government revenues means that government has to borrow for its other expenditures. This in turn increases the public debt which could result in debt distress and eventually into debt default, with potentially serious economic costs.

08.

(i)
- Productivity increases faster when countries produce goods and services in which they have a comparative advantage. Living standards can increase more rapidly.
- Global competition and cheap imports keep a constraint on prices, so inflation less likely to disrupt economic growth.
- An open economy promotes technological development and innovation, with ideas from abroad.
- Jobs in export industries tend to pay higher wages than jobs in import competing industries.
- Increase in government revenue due to the expansion of foreign trade which provides a reliable base for taxation.
- Consumers get access to goods and services which are not possible to produce locally
- Growing export sectors could be an inducement to attract foreign investors to the country

(ii)
- Protection of infant industries.
- Protection against dumping.
- Protection of strategic industries
- Protection of employment
- Diversification of domestic economy
- Source of government revenue
- Discouraging the consumption of demerit goods
- Developing self-sufficient national economy

(iii)
- Productivity
- Macroeconomic stability
- Infrastructure facilities
- Investment in R & D
- Tax structure
- Stable exchange rates
- Education system to develop human resources suitable to the current global situation
- Cost of doing business
  - Regulatory system
  - Flexibility in the labour market
  - Efficiency in public administration
  - Protection of private property
  - Rule of Law
  - Good governance

(iv)

(a) Balance of Trade:

Good = General Merchandise + Non-Monetary Gold

= (800 - 1200) + (5 - 55)

= -400 + (-50)

= Rs. - 450 million

(b) Balance on Goods and Services =

Goods + Services = -450 + Services

Services = Sea transport + Air transport + Travel + Financial services + Telecommunications

= (30 + 40 + 40 + (-5) + 45)

Services = 150

Balance on Goods and Services

= -450 + 150

= Rs. - 300 million

(c) Primary Income Account Balance:

Compensation of employees + Dividends + Interest income

= -5 + (-60) + (-35)

= Rs -100 million

(d) Secondary income Account Balance

Govt. Transfers + workers' Remittances

= 15 + 235 = Rs. 250 million

(v)

(a) An increase in real GDP:

With an increase in real GDP, the people of the relevant country will purchase more goods, including imports. As a result the supply of domestic currency for foreign currencies will increase. As the supply of domestic currencies against foreign currencies increases, the value of the domestic currency in terms of foreign currency depreciates.

(b) A decrease in the inflation rate:

If inflation rate decreases, a nation's exports become more competitive. As other nations buy these, now cheaper goods, the demand for domestic currency increases; there by driving up the exchange rate. Exchange rate appreciates.

Sub section 'B'

06.

(i) • Money is any commodity or token that is generally acceptable as a means of payment. A means of payment is a method of settling a debt. When a payment has been made, there is no remaining obligation between the parties to a transaction

- Under a barter system, individuals pay for goods or services by offering other goods and services in exchange. The problem with barter is that It requires a double - coincidence of wants
- Both traders must be willing to trade their products with each other. A situation that rarely occurs. Money solves the problem of double coincidence of wants because people will always money for the items they sell.
- Moreover barter is extremely expensive over long distances, What would it cost me to send rice to a distant place in return for an item available there. It is much cheaper to mail a cheque and get the item you want.
- Barter is time consuming because of difficulties in determining, the value of the product that is being -offered for barter.
- Difficulty in storage of the goods waiting for exchange
- Difficulty in engage in transactions which require smaller proportions of particular items

(ii) • A credit card is not money, but rather a mechanism for borrowing money, which must be repaid. It is just an ID card but one that lets you take out a loan at the instant you buy something. When you sign a credit card sales slip, you are saying " I agree to pay for these goods when the credit card company bills me". Credit card cannot be used as a store of value. It is only a temporary means of payment.

- To make the payment, you need money. You need to have currency or checking deposit to pay the credit card company. So although you use a credit card when you buy something, the credit card is not the means of payment and it is not money.

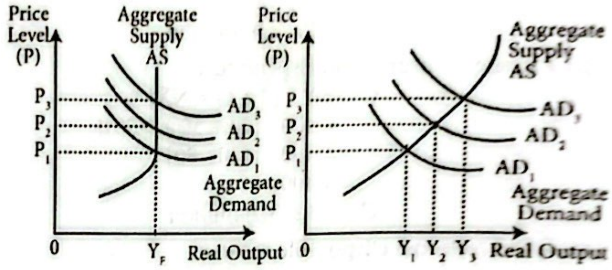
(iii) The macroeconomic variables that have the greatest effects on money demand are:

- real income
- interest rates
- the price level
- Expected rate of inflation

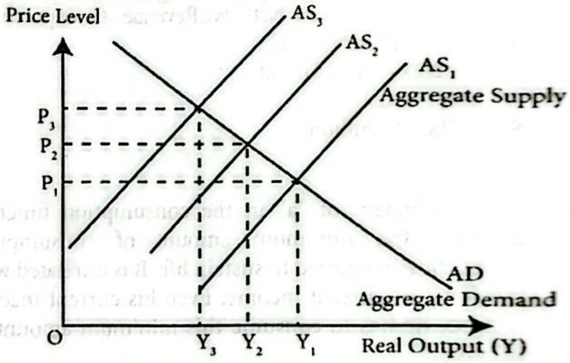
An important factor determining the number of transactions is real income. The amount of money demanded should increase when real income increases. Interest rates affect money demand through the expected return channel he higher the interest rate on money, the more money people will demand; however, the higher the interest rate paid on alternative assets to money, the more people want to switch from money to those alternative assets.

Higher the general level of prices, more rupees people need to conduct transactions and thus more rupees people want to hold~hen it is expected that the future rate of inflation to rise, the willingness to keep money balances at present reduces as the real value of money will fall in the future

(iv) Demand-pull inflation is the increase in the average price level caused by an excess of total spending, that is, aggregate demand. When firms cannot supply all the goods and services demanded by buyers, they respond by increasing prices. The average price level is thus pulled up by the total spending of buyers.



Cost-push inflation arises from events that increases the cost of production at each price level. With the increase in the cost of production, the aggregate supply curve shifts leftward while the aggregate demand curve remains unchanged.



(v) Increase in the net purchases of foreign currency by the Central Bank from the domestic foreign exchange market.

- Central Bank's SWAP arrangements with Bank of Ceylon and the National Savings Bank central Bank allowing them to borrow abroad in foreign exchange which is purchased by the Central Bank
- The government borrowing through sovereign loans which also enters into the money market as government spends the money.
- Central Bank's purchase of foreign currency obtained by the private sector by selling corporate debentures and equities to the foreign investors.
- Purchase of Treasury Bills by the Central Bank from the primary market.
- The reduction of the Statutory Reserve Ration by 200 basic points on 1st July 2013.
- Transfer of a part of Central Bank's profits to the government
- Reduced private sector borrowing from banks.

Eg: wages paid to labour services; cost of raw materials purchased; cost of transportation, electricity, advertising expenses etc.

Implicit costs are the costs that represent the value of resources used in production for which no monetary payment is made. They represent the imputed value that the firm's resources could command in their best alternative uses.

Examples:

- Implicit costs might include the foregone salary of the owner of the firm who used his/her time to run the business
- Forgone interest income because the owner invested his/her funds in the firm rather than depositing them in his savings account in the bank
- The forgone rental incomes due to sacrifice of land or buildings of the owner used for the firm
- Normal profit

(ii)
- A firm's owner often supplies entrepreneurial ability - the factor of production that organizes the business, makes business decisions, innovates, and bears the risk of running the business. The return to entrepreneurship is profit, and the minimum payment required-to keep the entrepreneur in this enterprise is called **normal profit**
- This normal return for performing entrepreneurial functions is an implicit cost. If his minimum or normal return is not realized, the entrepreneur will withdraw his efforts from this line of production and reallocate them to a more attractive line of production

(iii)
- Perfect competition (pure competition)
- Monopoly
- Monopolistic competition
- Oligopoly

(iv) Profit is the excess of total revenue over total cost of the firm. The firm will maximize profit or minimize losses by producing at that point where marginal revenue (MR) equals marginal cost (MC). For convenience we call this profit maximizing guide the **MR= MC rule**

The MR= MC rule can be conveniently restated in a slightly different form when being applied to a firm which acts as a price-taker. For a price-taker firm, product price is determined by the market forces of supply and demand and the firm cannot manipulate the price itself. The result is product price and marginal revenue are equal. So for a price-taker firm we may substitute the profit maximization rule as follows: To maximize profit or minimize losses the price-taker firm should produce at that point where price equals marginal cost (P = MC). This is simply a special case of MR= MC rule.

(v)

(a) Profit (or loss) = Total Revenue $(P \times Q)$ - Total Cost $(ATC \times Q)$

$= (50 \times 20) - (60 \times 20)$

$= 1\,000 - 1\,200 = -200$

Loss = Rs. 200

(b)

Total Fixed Cost	=	[Total Cost (ATC × Q)]	-	Total Variable Cost (AVC × Q)

$=(60 \times 20) - (40 \times 20)$

$=1200 - 800 =$ Rs. 400

Alternative answer:

Total Fixed Cost (TFC) = Average Fixed Cost (AFC) × Quantity of output

AFC = ATC - AVC = 60 - 40 = 20

TFC = AFC × Q = 20 × 20 = Rs.400

04.

(i)
- Output (product) approach
- Income approach
- Expenditure approach

Output, income and expenditure approaches to measuring aggregate economic activities of the economy reflect three different dimensions of the same transaction. Output approach measures .the value of the final goods and services produced in the country within a given year. Income approach measures the sum of all factor payments that were made to the factors who contributed to the production of all final goods and services. As the value of production should be equal to the total cost of the output, the value of total output and the value of total income are identical. Similarly, under the expenditure approach, the value of aggregate economic activity is measured by summing all the spending on final goods and services that has taken place throughout the year. Thus, we can determine the value of GDP for a particular year either by adding up all that was spent to buy total output or by adding up all the .money that was derived as income from its production. Buying and selling are two aspects of the same transaction.

(ii) **Private savings:**
- Household savings
- Business savings

} Domestic Savings

Government savings
- Net Factor Income from Abroad
- Net Private Current Transfers from Abroad

(iii) **Usefulness of National Income statistics:**
- Monitoring the performance of the economy over time.
- To measure economic growth
- International comparison of per capita income levels
- Analysis of the structure of the economy
- To assess standard of living and welfare of the people
- Identifying the pattern of functional income distribution
- Identifying the functional relationships between major macroeconomic variables
- Social and economic policy making
- Identifying the total resources of the economy and their utilization
- Analysing and forecasting the behavior of the macroeconomic variables of the economy

2015 Economics - I Answers

Que No.	Ans No.	Que No.	Ans No	Que No.	Ans No	Que No	Ans No	Que No	Ans No
01.	1	11.	3	21.	5	31.	4	41.	1
02.	2	12.	5	22.	4	32.	3	42.	1
03.	2	13.	4	23.	2	33.	3	43.	2
04.	4	14.	1	24.	5	34.	4	44.	1
05.	2	15.	1	25.	4	35.	3	45.	5
06.	5	16.	3	26.	2	36.	2	46.	2
07.	2	17.	2	27.	4	37.	4	47.	2
08.	5	18.	5	28.	4	38.	2	48.	1
09.	5	19.	4	29.	5	39.	5	49.	2
10.	1	20.	3	30.	5	40.	3	50.	3

2015 Economics - II Answers

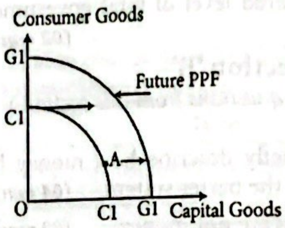
Sub section 'A'

01.

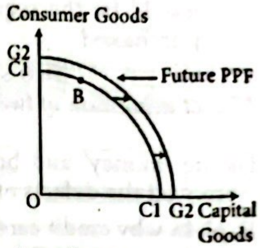
(i) **Free Good:** The goods which are unlimited in supply at- zero price.
Examples:- air,sunlight Or
A free good does not require any economic resources to produce it.
It is abundantly supplied at no cost.
Such goods are unlimited in supply and so have no opportunity cost.
Economic Good: The goods which are scarce in supply
Or
Economic goods require the use of scarce resources to produce them.
So their use does have an opportunity cost.]
Examples :- food, cloth, motor cars, computers etc.

(ii) The value of the next best alternative sacrificed whenever we make a choice in production or consumption is called opportunity cost.
Opportunity cost is subjective. Only the individual can estimate the expected value of the best alternative forgone.
The opportunity cost is essentially linked with the problem of scarcity. Opportunity cost exists because resources are scarce and they have alternative uses. Scarcity is the condition that results from the imbalance between unlimited wants and the limited resources / available for satisfying those wants. The fact that goods and services are scarce means that individuals cannot have all of everything they want. It is therefore necessary to choose among alternatives.
A sacrifice is involved whenever you make a choice One of the most important result of this fact is that whenever you make a choice, you must sacrifice or forgo other opportunities. The Opportunity not taken in making the decision is called the opportunity cost of the decision. When there is no scarcity there is no opportunity cost.

(iii) The future rate of economic growth of an economy is largely determined by the amount of current consumption sacrificed for current capital formation of the economy.
If the economy currently utilizes most of its resources in the production of capital goods, its productive capacity will increase and hence the future potential output will grow at a substantial rate. The future PPC will shift to the right in a major way
If the economy currently does not allocate a substantial amount of resources to the production of capital goods and spends most of its resources on current consumption, the growth of its future production possibilities will be very limited Even if the future PPC of this economy shifts to the right, it would be a marginal increase

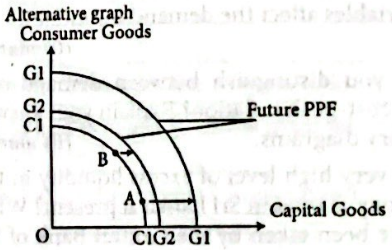


High Investment Economy



Low Investment Economy

Alternative graph



(iv) Economic efficiency means that the scarce resources of the economy should be used efficiently to produce goods and services most wanted by society.
Two types of efficiencies should be realized simultaneously to achieve economic efficiency - Productive efficiency and Allocative efficiency.
All points on the PPF are productively (technologically) efficient, meaning that it is impossible to produce more of one good without giving up some of the other. However, while productive efficiency is necessary for overall economic efficiency, so too is allocative efficiency, which identifies the specific point on the PPF that is the most valuable combination of outputs. Not all points on the PPF are allocatively efficient, meaning that there is a unique point that will be preferred over others since it best reflects wants. The combination of output which reflects the economic efficiency is deter mined by both marginal cost and marginal benefit of output. For this, in addition to the information provided by the PPF of marginal cost, the information on marginal benefits needs to be obtained from society's preference for goods and services.

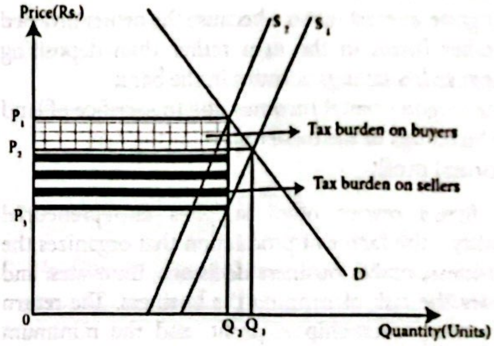
(v) A market economy is not necessarily to be a capitalist economy and a command economy too is not necessarily to be a socialist economy.
The division between capitalism and socialism is based on the ownership pattern of the productive resources of the economy. In a capitalist economic system, all productive resources are privately owned while in a socialist system resources are owned by the state.
On the other hand, the division between market economies and command economies is largely based on the resource allocation mechanism. In a market economy, resource allocation is determined by the price mechanism while in a command system, resource allocation is determined by centralized planning mechanism. This leads us to describe two extreme categories: market capitalism and command socialism. But this simple dichotomization raises the possibility of "cross forms", namely, market socialism and command capitalism. Although less common than the previous two, both have existed. Different combinations of resource ownership pattern and resource allocation mechanism have produced various forms of mixed economies. An important point to understand is that there are no pure examples of any type of system. All real economies are mixed economies exhibiting elements of various allocation and ownership systems

02.
(i) - Prices of related goods, - Expected future prices
- Technology, - Number of suppliers
- Government policies: Taxes and subsidies
- Prices of resources used to produce the good

(ii) When the price of a good rises, the other things remain the same, its real income falls. As real income falls people buy less of that good if it is a normal good. Similarly, when the price of a good falls and all other factors remain constant, the real income rises. As real income rises, people buy more of that good if it is A normal good. Thus the income effect supports the Law of Demand.
When the price of a good rises, the other things remain the same, its relative prices rises. As the relative price of that good rises, people buy less of that good, and more of its substitutes. Similarly, when the price of a good falls, the other things remaining the same, its relative price falls. As the relative price of that good falls, people buy more of that good, and less of its substitutes. Thus the substitution effect also supports the Law of Demand.

(iii) Price floors are minimum prices set by the government for certain commodities that it believes are being sold in an unfair market with too low of a price and thus their producers deserve some assistance.
Eg: Fixing minimum wages
Introducing guaranteed prices for agricultural crops
A price ceiling is a minimum price set by the government for particular goods and Services that they believe are being sold at too high of a price and thus consumers need some help purchasing them.
Eg: -Fixing maximum prices for some essential consumer items such as rice, milk food, bread, passenger transport by bus, LP gas,
- Fixing maximum prices for interest rates
- Rent control

(iv) As a rule, the side of the market that is less elastic bears a higher burden of the tax. Since supply is more inelastic, producers bear a much higher share of the tax burden.

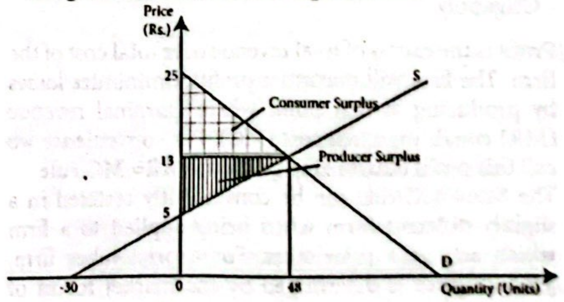


(v) (a)

$Qd = Qs$
$100 - 4P = -30 + 6P$
$100 + 30 = 4P + 6P$
$130 = 10P$
$P = 13$

$Qd = 100 - 4P$	$Qs = -30 + 6P$
$= 100 - 4 \times 13$	$= -30 + 6 \times 13$
$= 100 - 52$	$= -30 + 78$
$Q = 48$	$Q = 48$

Diagram to show market equilibrium



(b) Producer surplus and Consumer surplus:
Producer surplus

$$= \frac{13 - 5}{2} \times 48 = \frac{384}{2} = 192 /=$$

Consumer surplus

$$= \frac{25 - 13}{2} \times 48 = \frac{576}{2} = 288 /=$$

03.
(i) Explicit costs are the direct cost of resources that the firm uses that are supplied outside the firm.
Alternative answer:
Explicit cost is the opportunity cost of inputs which the firm purchases from outside.

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(iv)

(a) **Net exports:**

Current Account Balance = Net Exports + Net Foreign Factor Income Payments + Net Foreign Current Transfers

Net Exports = Current Account Balance - (Net Foreign factor Income + Net Foreign Current Transfers)

= -25 - [(7 - 12) + 10]

= -25 - [-5 +10]

= -25 - 5

= Rs. - 30 million

(b) **Private Consumption Expenditure**

C = Y - [I + G + NX]

= 400 - [70 + 50 + (-30)]

= 400 - [120 - 30]

= 400 - 90

C = Rs. 310 million

(c) **Gross National Disposable Income**

= GNP+ NFI (=Net factor incomes from abroad) + Net Foreign Current Transfers

= 400 + (-5) + 10

= Rs. 405 million

(d) **Government Savings (S_G)**

= Net Govt. Revenue - Govt. purchases

S_G = (T - Tr - INT) - G

= (100 - 25 - 15) - 50

= 60 - 50

S_G = Rs. 10 million

05.

(i) **The coefficient of "a" in the consumption function denotes the minimum amount of consumption expenditure required to sustain life. It is unrelated with the level of current income. Even his current income is zero, he has to consume this minimum amount to survive.**

The coefficient "b" is the marginal propensity to consume (MPC), or the fraction of an increase (change) in income that is consumed. ($\Delta C / \Delta Y$)

(ii) **National income is said to be in equilibrium when there is no tendency for it either to increase or to decrease. The particular value of national income that exists in equilibrium is called the equilibrium national income. Aggregate output must equal aggregate expenditure in an open economy:** Y = E

Y = C + I + G + (X - M)

Total leakages equal total injections in an open economy W = J

S + T + M = I + G + X

(iii) **In a closed economy when the planned saving is greater than the planned investment (that is S > I), aggregate planned expenditure will be less than the aggregate output (Y>E). This will result a process of convergence towards equilibrium aggregate expenditure**

As the output level exceeds total expenditure, there will be an increase in inventories. When inventories increase above the target level, firms cut down the production to restore inventories to the target level. Consequently, the real GDP fall until it reaches the equilibrium level at which planned savings equal planned investment (S=I). When aggregate planned expenditure equals real GDP, there are no unplanned inventory changes and real GDP remains constant at equilibrium expenditure level.

(iv)

(a) **Equilibrium level of National Income**

Y = C + I + G

Y = 50 + 0.8 Yd + 120 + 100

Y = 50 + 0.8 (Y - T + Tr) + 220

Y = 50 + 0.8 (Y - 80 + 55) + 220

Y = 50 + 0.8y - 64 + 44 + 220

Y = 250 + 0.8y

Y = - 0.8y = 250

Y = (1 - 0.8) = 250

0.2 y = 250

Y =Rs. 1250 million

(b) **New Equilibrium Income Level when G = 200**

Y = 50 + 0.8yd + 120 + 200

Y = 50 + 0.8 (y - 80 + 55) + 320

Y = 50 + 0.8y - 64 + 44 + 320

Y = 350 - 0.8y

Y - 0.8y = 350

0.2y = 350

Y = 350/0.2 = 1750 billion

Alternative answer:

$K_G = \frac{1}{1-b} = \frac{1}{1-0.8} = \frac{1}{0.2} = 5$ (Govt. Expenditure Multiplier)

$\Delta G = 100 \rightarrow \Delta Y = \Delta G \times KG = 100 \times 5 = \Delta Y = 500$

Therefore, new equilibrium level of income = 1250 + 500 = 1750 billion

(c) **Govt. Expenditure multiplier:**

$K_G = \frac{1}{1 - mpc} = \frac{1}{1 - 0.8} = \frac{1}{0.2} = 5$

(d) **To achieve target level of income 2000:**

Current level of income = 1750 million

Target Level = 2000 million

Difference of Income Level = ΔY = 250 million

If the Govt. expenditure multiplier is 5, to achieve an additional income of Rs. 250 million, the additional govt. expenditure required is: 250/5 = Rs. 50 million. Therefore, to achieve income level of Rs. 2000 million, the required level of total government expenditure is Rs. 250 million (= 200 + 50)

Alternative answer:

Y = C + I + G

2000 = 50 + 0.8 Yd + 120 + G

2000 = 50 + 0.8 (y - T + Tr) + 120 + G

2000 = 50 + 0.8 (y - 80 + 55) + 120 + G

2000 = 50 + 0.8y - 64 + 44 + 120 + G

2000 = 150 + 0.8(2000) + G

2000 = 150 + 1600 + G

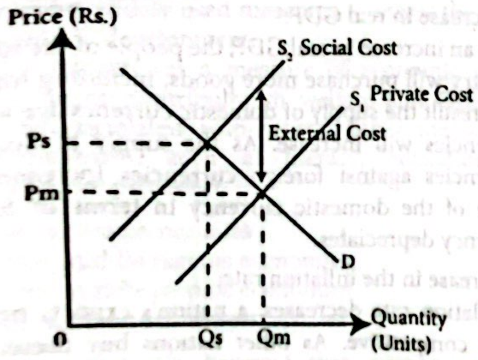
2000 - 1750 = G = 250

Measures taken to absorb excess liquidity:
- **Open market operations throughout the years**
- **Overnight repurchase agreements**
- **Introduction of Standing Deposit Facility (SDF)**
- **As the CB's holding of govt. Securities were relatively low, Treasury bills borrowed from the Employees Provident Fund , were also used to manage excess liquidity.**
- **Encouraging commercial banks to utilize their ample liquidity to channel credit flow for productive economic activity.**
- **CB placed a limit of three times per calendar month on the use of the SDF by OMO participants.**

17.

(i)
- **A pure public good is nonexcludable. It is impossible (or extremely costly) to prevent someone from benefiting from it. Once it is produced it is available to all and it can be consumed collectively.**
- **A pure public good is non rival in consumption, If its use by one person does not decrease the quantity available for someone else.**
- **Merit goods are those goods and services consumption of which generates positive externalities where the social benefit from consumption exceeds the private benefits.**
- **When these goods and services are provided through market forces, there is a tendency for under-consumption.** eg. education, health care

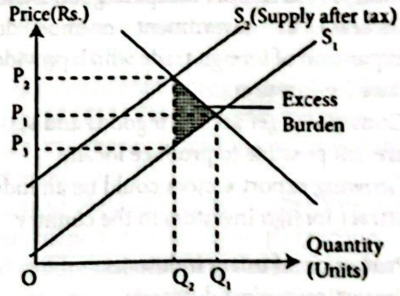
(ii)
- **When the production of a good or service entails external costs, market failure occurs because the firm fails to take into account the external costs .**
- **When there are external costs in production, the social cost exceeds the private cost. But the market decisions are based, pure on private costs and benefits. As a result, too much output is produced at too low a price**
- **There is an over - allocation of scarce resources in the production of this good. When there are external costs, the equilibrium market output level will exceed the socially optimum quantity. It is an inefficiency in resource allocation.**



- **When external costs are not taken into account, the market equilibrium output is Q_m in contrast to the socially optimum output level of Q_s**

(iii)- A proportional tax is one whose average rate remains the same regardless of the size of the income. The tax collected is in constant proportion to the income being taxed
- A progressive tax is one whose average rate increases as income increases. More well-off are taxed at a higher rate than are the less well-off
- A regressive tax is one whose average rate declines as income increases. One pays in tax a smaller and smaller proportion of one's income as income increases, though it is possible for the absolute amount to increase

(iv)
- Taxes generally impose an excess burden on taxpayers, a cost beyond the amount of money that he or she hands over to the tax collector. Excess burden is caused by tax induced distortions in household and firm behavior.
- Taxes cause households or firms to shift their economic choices, such as choose among different goods, inputs, locations etc. The total economic burden of a tax includes both payments that taxpayers make to the government and any lost economic value from inefficient activities undertaken in reaction to taxes. The excess burden of taxation is the magnitude of the economic costs of accompanying economic distortions. For example, a tax on labor income typically discourages work by encouraging inefficient substitution of untaxed leisure for taxed paid work.
- Excess burden, therefore, is the loss of economic surplus belongs to consumers and producers, as a consequence of distortions in economic decision making resulting from taxes. The loss of welfare is an addition to the financial burden of taxation.



(v)
- Sri Lanka's public debt is one of the highest among the emerging economies. The size of the total public debt is often expressed as a percentage of GDP. This does not clearly reflect the real situation of the debt burden. For example, the figures given below indicates that over the recent years, the total public debt as a percentage of -GDP has continued to fall.
  2010 - 81.9% , 2011 - 78.5% , 2012 - 79.2% , 2013 - 78.3%
- By looking at these figures someone may conclude that the trend is more favorable to the country.
  But in reality, Sri Lanka is currently facing a serious situation with respect to the servicing of public debt. Comparing the ratio of government debt